

Mohamed Zayed Ahmed

+20 128 490 7633 — moh.z.ahmed007@gmail.com — linkedin.com/in/mozayed007 — github.com/mozayed007

Professional Summary

AI Research Engineer with 3+ years of experience architecting and productionizing multi-tenant AI platforms, agentic workflows, RAG systems, and high-throughput distributed ML pipelines. Proven track record at Sprints AI and Dark Entry delivering production FastAPI services, hybrid vector/graph retrieval (Qdrant/HelixDB), neuroscience-inspired memory systems, and 3× throughput gains in data processing engines. Strong research foundation in generative models (T5 for antigen-antibody interactions surpassing SoTA by 10%), multi-agent systems, Bayesian optimization for pharma processes, and LLM evaluation frameworks. Passionate about building scalable, responsible AI systems for education, scientific discovery, and real-world applications.

Experience

Sprints AI

Research Engineer - R&D

Cairo, EG

Sep 2025 – Present

- Architected and implemented a multi-tenant AI Learning Companion platform with production FastAPI services, org-scoped authentication, Supabase/Postgres persistence, Qdrant retrieval, and Redis-backed state management.
- Built a precision content-ingestion pipeline for videos, PDFs, code, markdown, and curriculum maps, preserving timestamps, spatial coordinates, line ranges, and hierarchy metadata to support exact citations and navigation.
- Developed production agent workflows for tutoring, career coaching, study tools, and educational games, combining persona-based orchestration, curriculum-aware context, hybrid retrieval, and backend-switchable Qdrant/HelixDB search.
- Implemented neuroscience-inspired memory, guardrails, and resource-safety layers for personalized learning flows, including episodic/semantic/procedural memory, preference recall, and VPS-safe ingestion under constrained infrastructure.
- Productionized HelixDB v2's graph+vector hybrid retrieval within the Learning Companion, achieving 50% fewer DB round-trips and superior curriculum-aware relevance.
- Built a dual RAG provider adapter (Qdrant ↔ HelixDB) with runtime switching and graceful fallback, enabling zero-downtime provider changes.

Dark Entry

Research Engineer (Contractor)

Remote

Jun 2025 – Dec 2025

- Architected production platforms and ingestion orchestrators handling 50+ concurrent feeds, featuring custom adapters, SHA256 fingerprinting, and Redis-backed queuing.
- Developed a dual-implementation (Python/Rust) data processing engine using Rayon and Crossbeam for 3× throughput gains via tiered pattern matching.
- Built a health-tracking system with configurable cooldowns and self-healing auto-recovery logic, reducing manual intervention by 80%.
- Implemented distributed system patterns including Circuit Breaker state machines, adaptive rate limiters, and semaphore-based concurrency control for external APIs.
- Refactored monolithic services into modular FastAPI architectures, reducing database queries by 70% via bulk operations and NDJSON streaming for multi-terabyte datasets.
- Hardened infrastructure with Docker/Compose, Celery/Redis task orchestration, and structured observability using Prometheus metrics and OpenTelemetry tracing.

Freelance Consulting & Contracts / Self-Employed

ML/AI Engineer - Consultant & Contractor

Remote

Sep 2023 – Present

- Contracted with multiple startups, businesses, and research groups, delivering innovative ML-engineering and R&D solutions across education, pharma, and NLP domains.
- Architected a multi-agent research framework using MCP servers and sequential pipelines to orchestrate autonomous information gathering and synthesis workflows (DeepResearch-inspired).
- Designed and built a responsible, end-to-end autonomous NER pipeline for unstructured data using spaCy and transformer models, featuring continuous fine-tuning via active learning, responsible AI guardrails, and automated ingestion.
- Developed and benchmarked oblique decision tree (ODT) controllers for pharmaceutical process control, extracting explicit control laws and comparing learned decision regions against MPC baselines.
- Built a pharmaceutical process optimization framework using GP, DKL, BART, and CVAE models for experiment selection, design-space analysis, and inverse recipe generation.
- Implemented a multi-process async benchmark system (for Stakpak) to evaluate and optimize LLMs, boosting experimentation velocity by $3\times$ (from 30-50 min to 10 min) and enabling seamless provider switching across OpenAI, Anthropic, Google, and open-source models.
- Explainable dynamic modeling: Designed scalable control strategies for life science processes under uncertainty using Bayesian methods (MOBO, GP, BART) and generative models (CVAE).

Superprof

Online Tutor · Freelance

Remote (United States)

Aug 2022 – Present

- Tutored 350+ hours of personalized online sessions for undergraduate (B.Sc.) and graduate (M.Sc.) students from diverse backgrounds in Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Generative AI, and Cybersecurity.
- Improved student performance on advanced SOTA topics and capstone projects through structured mentoring and project-based learning.

Proteinea

Deep Learning Intern

Cairo, EG

Aug 2022 – Nov 2022

- Developed initial T5-based models for antigen-antibody interactions, laying the foundation for a graduation thesis that later surpassed 2022 SoTA baselines on the SABDAB dataset.
- Automated evaluation pipelines and created TensorBoard visualizations, reducing analysis time by 30%.
- Researched bio-informatics topic (Protein-Protein interactions & generation) for a sponsored graduation project.

Projects

Antigen-Antibody Translation/Generation – PyTorch, Hugging Face, T5, Pandas, AWS Graduation Research Thesis

- Developed a custom T5 Transformer that achieved 10% higher performance than 2022 SoTA baselines on the SABDAB dataset using AWS NVIDIA A10G GPUs (building on Proteinea internship insights).
- Enabled real-time monitoring with custom TensorBoard dashboards and facilitated team collaboration using Termius Vault and Tmux.
- Showcased robust cloud-based optimization capabilities for protein-protein interaction prediction in bioinformatics.

Multi-Agent Research System – Google ADK, Pydantic, Asyncio, FastAPI, Gradio

- Migrated a research workflow to Google's Agent Development Kit (ADK), creating a SequentialAgent pipeline for data collection, analysis, and report synthesis.
- Integrated external APIs (BraveSearch, ArXiv, Wikipedia) and SQLite persistence to enhance system modularity and data gathering capabilities.
- Drafted architecture diagrams and communication patterns to streamline development and ensure scalability.

LiveMem / THEN – PyTorch, nanochat, NumPy, Attention Hooks

Independent Research

- Architected a stateful “live memory” extension for nanochat (THEN architecture) that decouples weight training from post-training episodic memory ingestion and query-time recall.
- Implemented a disk-tiered memory manager using numpy.memmap and chunked top-k retrieval to stream long-horizon traces from disk with bounded device-memory usage.
- Built ingestion and query pipelines for frozen-model memory population and stateful recall, including persistence validation and synthetic long-horizon episode generation for benchmarking.

LLM Evaluation System – PyTorch, Asyncio, OpenAI, Anthropic

Stakpak, Software Engineering

- Developed an LLM benchmark system to align and optimize domain-specific responses using multi-processed asynchronous generation, achieving a 300% speed increase.
- Designed an abstract generation & evaluation system supporting multiple LLM providers, ensuring interoperability and high performance for real-world NLP applications.

Blossom AI: Image Classifier with Transfer Learning – PyTorch, Timm, Torch-Vision *Computer Vision Coursework*

- Developed an image classifier using PyTorch with pre-trained models, achieving 94% testing accuracy on the Flower-102 Dataset with reduced computing costs through parameter-efficient fine-tuning.
- Optimized vision models (EfficientNet, CCT) using LoRA for fine-tuning and QLoRA for quantization, achieving approximately 40-60% reduction in compute costs while fine-tuning around 50% of the model parameters.

Media Recommendation System – Milvus, PyTorch, LitServe, Pandas, NumPy, NLTK

- Built an ETL pipeline using MAL RESTful API v2 for MyAnimeList database ingestion.
- Classical content-based filtering recommendation system based on categorical/numerical features.
- Developed an NLP-powered semantic similarity engine using embedding models and a vector database (Milvus), enhancing recommendation relevance through context-based synopsis analysis.

Skills

Programming Languages: Python, Rust, C++, SQL, Bash, PowerShell scripting.

Frameworks & Libraries: PyTorch, TensorFlow, FastAPI, LitServe, Triton, SQLAlchemy, Celery, Redis, Scrapy, Selenium, Playwright, Pydantic, Pandas, NumPy, Hugging Face (Transformers, PEFT), Scikit-Learn, Timm, OpenCV, NLTK, LangChain, LlamaIndex, OpenAI, Anthropic, Together, PySpark, structlog, bun, NextJS.

Domain Expertise: Distributed Systems & System Architecture, Machine Learning & Deep Learning, NLP (NER, LLMs, RAG, Agentic Systems), Computer Vision, Generative AI, Statistical Inference & Data Analysis, Vector Databases (Qdrant, Milvus, Helix, Weaviate), Bayesian Optimization & Explainable AI.

Tools & Platforms: Git, Docker, Docker Compose, AWS, GCP, Azure, Ollama, Linux, WSL, SQLite, PostgreSQL, REST APIs, Kaggle, Jupyter Notebooks, Prometheus, OpenTelemetry, Termius, Tmux.

Education

University of Science and Technology – Zewail City

Giza, EG

Bachelor of Science, Communication and Information Engineering

Jun 2024

- **Concentration Courses:** Statistical Inference and Data Analysis, Signal Processing, Digital Signal Processing, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision.

Additional Information

Languages: English (Fluent), Arabic (Native)

Interests: Neuroscience-inspired AI, Multi-Agent Systems, Responsible AI, AI for Scientific Discovery, Open-Source Contributions